

MITE 435: Software Design Pattern

Lab Work Scenario

Suppose IIT recently launched EduSphere, a learning management system that every department uses to run its courses. When a teacher creates a new course in EduSphere, the system automatically sets up two essential components for that course: a Lecture component and an Assessment component. The university offers courses in two delivery modes, Online and Offline, and each mode handles lectures and assessments in its own way.

The Lecture component is responsible for two things: delivering the class and sharing study materials.

In Online mode, the teacher conducts the class through a live video session. Students join from their laptops or phones, and the teacher shares slides on screen. After the class, the recording and slides are uploaded to the course page on the portal so students can review them anytime. The online lecture prints "Delivering lecture via live video session" when delivering, and "Uploading recording and slides to the course portal" when sharing materials.

In Offline mode, the teacher conducts the class face to face in a classroom using a whiteboard and projector. After the class, printed handouts are distributed to the students present, and extra copies are kept at the department office for anyone who missed the class. The offline lecture prints "Delivering lecture in the classroom" when delivering, and "Distributing printed handouts to students" when sharing materials.

The Assessment component is responsible for two things: conducting the assessment and publishing results.

In Online mode, students take a timed MCQ quiz on the online portal. The portal grades the quiz automatically as soon as it is submitted, and each student can see their marks on their dashboard right away. The online assessment prints "Conducting MCQ quiz on the online portal" when conducting, and "Publishing auto-graded results on student dashboards" when publishing results.

In Offline mode, students sit a written exam on paper in the exam hall under the supervision of an invigilator. The teacher collects the scripts, grades them manually, and the results are posted on the department notice board. The offline assessment prints "Conducting written exam on paper" when conducting, and "Posting manually graded results on the notice board" when publishing results.

You have been hired as a developer on the EduSphere team. Your job is to design the part of the system that creates the lecture and assessment for a course so that these requirements are met.

Your Task

Design and implement this system using the Abstract Factory pattern. You may use any object-oriented language, but Java is preferable.